

Research has multifaceted benefits for the economy and society at large. It shapes academic culture, informs teaching, and advances knowledge by filling gaps and identifying new areas of study. However, incorporating AI into research practices brings a new level of complexity and challenges to established norms. While AI can relieve researchers and institutions of tedious tasks and improve efficiency in a competitive academic landscape, concerns arise over potential biases introduced in AI-assisted peer review, limitations in identifying nuance and novelty, and the risk of using AI for surveillance and algorithmic management. Expanding on Beer's concept of "cultural speedup" resulting from data analytics and AI algorithms, we consider the implications for research creativity. As the UK government targets substantial investments in R&D and AI research continues to grow rapidly, it is imperative that responsible innovation and ethical considerations are integrated into research practices, culture, and institutional governance.

The impact of AI and digital technologies on research and science culture is still in the early stages of research. While AI tools are being developed to assist with various aspects of academic life, such as selecting reviewers, checking the efficacy of papers, summarizing findings, and identifying plagiarism, their use in peer review is still debated due to concerns over bias. AI is also viewed as a "risky fix" and a "runaway process" in science, with potential unforeseen outcomes that could reshape knowledge production. Despite the concerns, the increased use of AI in science could provide a range of benefits, such as enabling

